Bastián Esteban Castorene Castorene

bastian.castorene@gmail.com   @bastiancastorene   Date of birth: 21/08/2000

Education

2024 – present — **PhD Candidate in Physics, Joint Program between Universidad Técnica Federico Santa María (UTFSM) and Pontificia Universidad Católica de Valparaíso (PUCV).**
Research focus: Quantum thermodynamics and magnetic systems.
Thesis: *Qubit-based quantum heat engines*
Funded by the ANID National Doctoral Fellowship (No. 21250015).

2020 – 2023 — **Bachelor of Science in Physics, Universidad Técnica Federico Santa María (UTFSM).**
Graduated *Summa Cum Laude*.
Thesis: *Otto and Stirling Engines in a System of Three Entangled Qubits in a Ring Topology.*

Scientific Publications and Contributions

Articles

1. B. Castorene, M. H. Groves, F. J. Peña, et al. “Caloric phenomena and Stirling-cycle performance in Heisenberg-Kitaev magnon systems”. In: *Phys. Rev. E* 114 (1 July 2026), p. 014132. DOI: 10.1103/msbk-z1mc.
2. B. Castorene, F. J. Peña, M. H. Groves, and P. Vargas. “Exact Combinatorial Density of States for the Critical 1D Ising Model”. In: *Entropy* 28.7 (2026). ISSN: 1099-4300. DOI: 10.3390/e28070821.
3. B. Castorene, F. J. Peña, E. Suárez Morell, C. Lewenkopf, M. H. Groves, N. Cortés, and P. Vargas. “Reaching maximum efficiency in quantum Stirling engines using multilayer graphene”. In: *Phys. Rev. E* 113 (4 Apr. 2026), p. 044126. DOI: 10.1103/s59t-ygkq.
4. N. Cortés, B. Castorene, F. J. Peña, D. Melo, S. E. Ulloa, and P. Vargas. “Thermal Hofstadter Butterflies”. In: *Nano Letters* (July 2026). DOI: 10.1021/acs.nanolett.6c01000.
5. B. Castorene, V. G. de Paula, F. J. Peña, C. Cruz, M. Reis, and P. Vargas. “Quantum Level-Crossing Induced by Anisotropy in Spin-1 Heisenberg Dimers: Applications to Quantum Stirling Engines”. In: *Advanced Quantum Technologies* 8.10 (2025), e2500204. DOI: https://doi.org/10.1002/qute.202500204.
6. B. Castorene, F. J. Peña, A. Norambuena, S. E. Ulloa, C. Araya, and P. Vargas. “Entropy, entanglement, and susceptibility of three qubits near quantum criticality”. In: *Physical Review E* 111.3 (2025), p. 034118. DOI: 10.1103/PhysRevE.111.034118.
7. F. J. Peña, C. D. Nuñez, B. Castorene, M. Aguilera, N. Cortés, and P. Vargas. “Ratio between Seebeck coefficient and entropy per particle as a tool for elementary charge determination”. In: *Phys. Rev. E* 112 (6 Dec. 2025), p. 065508. DOI: 10.1103/rq83-rwk2.
8. B. Castorene, F. J. Peña, A. Norambuena, S. E. Ulloa, C. Araya, and P. Vargas. “Effects of magnetic anisotropy on three-qubit antiferromagnetic thermal machines”. In: *Physical Review E* 110.4 (2024), p. 044135. DOI: 10.1103/PhysRevE.110.044135.
9. C. Araya, F. J. Peña, A. Norambuena, B. Castorene, and P. Vargas. “Magnetic Stirling Cycle for Qubits with Anisotropy near the Quantum Critical Point”. In: *Technologies* 11.6 (2023), p. 169. DOI: 10.3390/technologies11060169.

Manuscripts

1. B. Castorene, M. H. Groves, F. J. Peña, E. E. Vogel, and P. Vargas. “Fundamental Work Scaling and Non-Extensivity in Critical Quantum Stirling Engines”. 2026. arXiv: 2510.25533 [cond-mat.stat-mech].

Conferences and Presentations

Oral Presentations

June 2026 — **Talk at Panoramas 2026, Pontificia Universidad Católica de Valparaíso, Campus Curauma, Chile.**
Fundamental Work Scaling and Non-Extensivity in Critical Quantum Stirling Engines.

May 2026 — **Talk at XXIII TREFEMAC 2026, Centro Atómico Bariloche-Instituto Balseiro, Argentina.**
Fundamental Work Scaling and Non-Extensivity in Critical Quantum Stirling Engines.

Conferences and Presentations (continued)

Mar 2026 — **Talk at MARCh 2026, Universidad Técnica Federico Santa Maria, Valparaíso, Chile.**
Qubit-based quantum heat engines: Maximum quantum work at criticality.

Dec 2025 — **Talk at the IV International Workshop on Statistical Physics, Antofagasta, Chile.**
Exact Conditions for Carnot Efficiency in Quantum Stirling Engines at Criticality.

Talk at the 20th International Seminar on Condensed Matter Physics and Statistical Physics (SIMAFE 2025), Temuco, Chile.
On the Maximum Quantum Work in Critical Stirling Engines: Applications to Fibonacci–Lucas Degeneracies.

Nov 2025 — **Talk at the VIII National Congress on Nanotechnology, La Serena, Chile.**
Qubit vs. Qutrit: Efficiency and Entanglement in Stirling Cycles.

Oct 2025 — **Talk at XXIII B-MRS Meeting, Salvador de Bahía, Brazil.**
Quantum Level-Crossing Induced by Anisotropy in Spin-1 Heisenberg Dimers: Applications to Quantum Stirling Engines.

Sep 2025 — **Co-speaker at Café de Investigación V20, Departamento de Ingeniería en Diseño, UTFSM, Valparaíso, Chile.**
Quantum Thermal Machines in Two-Dimensional Materials, with Dr. Francisco J. Peña. Link

Jun 2025 — **Co-speaker at DFI Seminar, Universidad de Chile, Santiago.**
Quantum Thermodynamic Cycles: Otto & Stirling, with Dr. Francisco J. Peña.

May 2025 — **Talk at Panoramas 2025, Pontificia Universidad Católica de Valparaíso, Campus Curauma, Chile.**
Spin-1/2 vs. Spin-1: A Quantum Thermodynamic Study of Anisotropic XXX Heisenberg Models. Link

Apr 2025 — **Talk at IV QUSantiago, Universidad de Santiago, Santiago, Chile.**
Quantum Thermodynamic Cycles: Stirling & Otto. Link

Mar 2025 — **Talk at MARCh 2025, Universidad de Mendoza, Mendoza, Argentina.**
Spin-1/2 vs Spin-1: Comparisons of Stirling Engines.

Jan 2025 — **Talk at XIII Escuela de Nanoestructuras, UTFSM, Campus Casa Central, Chile.**
Spin-1/2 vs Spin-1: A Quantum Thermodynamic Study.

Oct 2024 — **Final Presenter at SOCHIFI 2024, Universidad de la Frontera, Temuco, Chile.**
Reduced Entropy and the Effective Temperature of a Three-Qubit System.

May 2024 — **Talk at Panoramas 2024, Pontificia Universidad Católica de Valparaíso, Campus Curauma, Chile.**
Effects of Magnetic Anisotropy on Three-Qubit Antiferromagnetic Thermal Machines. Link

Jan 2024 — **Talk at XII Escuela de Nanoestructuras, UTFSM, Campus Casa Central, Chile.**
Otto and Stirling Engines in a System of Three Entangled Qubits in a Ring Topology.

Oct 2023 — **Talk at LAW3M, Puerto Varas, Chile.**
Magnetic Stirling Cycle for Qubits with Anisotropy near the Quantum Critical Point.

Conferences Attended

May 2026 — **Postgraduate Course: Active Matter and Non-equilibrium Phenomena**, Centro Atómico Bariloche-Instituto Balseiro, Argentina.
Intensive course attended prior to the XXIII TREFEMAC congress.

Apr 2026 — **Attendee at V QuSantiago 2026**, Universidad Diego Portales, Santiago, Chile.
V edition of the quantum technologies conference (Apr 14–16).

Jan 2026 — **Attendee at Congreso Futuro 2026: Humanidad, ¿Hacia Dónde Vamos?**, Centro Cultural CEINA, Santiago, Chile.
Full participation in the six-day program (Jan 12–17), featuring over 120 international speakers on science, technology, and humanities.

Nov 2025 — **Invited Attendee at Science 2030: Sowing Innovation, Projecting the Future**, Viña del Mar, Chile.

Conferences and Presentations (continued)

- **Attendee at the IX Quantum Optics and Solid State Conference**, Valparaíso, Chile.

Academic Service & Organization

Apr 2026 – Present
- **Representative of the PhD Students** , Joint PhD Program in Physics, UTFSM & PUCV.
  Elected to represent the student body of the joint doctoral program across both institutions, coordinating student affairs and academic initiatives.

Mar 2026
- **Scientific & Organizing Committee Member**, MARCh 2026, Universidad Técnica Federico Santa Maria, Valparaiso, Chile.
  Oversaw the conference digital infrastructure and abstract submission pipeline. Orchestrated the scientific peer-review process and curated the final academic program.

May 2024
- **Chair & Lead Organizer**, Panoramas 2024, Pontificia Universidad Católica de Valparaíso, Campus Curauma, Chile.
  Spearheaded the event organization, managing comprehensive logistics, speaker scheduling, and the selection of scientific contributions.

Jan 2024
- **Local Organizing Staff**, XII Escuela de Nanoestructuras, UTFSM, Campus Casa Central, Chile.
  Facilitated on-site operations and provided audiovisual support for speakers, ensuring the seamless execution of scientific sessions.

Awards, Fellowships and Distinctions

Grants

Mar 2025 – Present
- **ANID National Doctoral Fellowship (No. 21250015).** Fellowship awarded for doctoral studies in Physics in Chile. Link

Jun 2026 – Mar 2027
- **Scientific Research Initiation Program (PIIC), Universidad Técnica Federico Santa María.**
  Project No. 005/2026: *Non-extensivity in quantum systems with applications in Stirling Engines.*

2026
- **UTFSM Research Internship Travel Grant.**
  Awarded for participation in the *Research Internship Program*, Universitat Autònoma de Barcelona, Spain.

Jun 2025 – Mar 2026
- **Scientific Research Initiation Program (PIIC), Universidad Técnica Federico Santa María.**
  Project No. 004/2025: *Magnetic Textures and Their Role in Thermodynamic Engines.*

Dec 2025
- **UTFSM Conference Travel Grant.**
  Awarded for participation in the *IV International Workshop on Statistical Physics*, Antofagasta, Chile.

Nov 2025
- **National Nanotechnology Congress Grant.**
  Awarded by the *Ciencia 2030 Consortium* (UTFSM–Universidad de Valparaíso) to support participation in the *VIII National Congress on Nanotechnology*, La Serena, Chile.

Oct 2025
- **UTFSM Conference Travel Grant.**
  Awarded for participation in the *XXIII B-MRS Meeting*, Salvador de Bahía, Brazil.

Jul 2025
- **InnovaCiencia International Internship Fellowship.** Research internship at Phasecraft, London, July 2025.

Mar 2024 – Mar 2025
- **UTFSM Internal Doctoral Fellowship.** Agreement No. 037/2024: Tuition and financial support for doctoral studies in Physics. Link

Distinctions

2025
- **Wolfram Research Grant.** Awarded a research license for Wolfram Mathematica in recognition of the article *Entropy, Entanglement, and Susceptibility of Three Qubits near Quantum Criticality* (Phys. Rev. E 111, 034118, 2025).

2021
- **Bronze Medal, 2021 University Physics Competition.** Team 120 under faculty sponsor Dr. Marat Siddikov (UTFSM). Link

Research Experience

Mar 2025 – Present	**Research Technician**, Fondecyt Project No. 1250173. Principal Investigator: Dr. Francisco J. Peña.
May 2024 – Present	**PhD Thesis Student**, Fondecyt Project No. 1240582. Principal Investigator: Prof. Patricio Vargas.
Aug 2023 – Mar 2025	**Research Technician**, Fondecyt Initiation Project No. 11221088. Principal Investigator: Dr. Natalia Cortés.
Jun 2022 – Mar 2023	**Research Technician**, Fondecyt Regular Project No. 1221827. Principal Investigator: Dr. Taisiya Mineeva.

Teaching Experience

Mar 2025 – Jul 2025	**Teaching Assistant**, UTFSM. Course: PhD Program FIS330, Statistical Mechanics.
Aug 2024 – Dec 2024	**Teaching Assistant**, UTFSM. Course: PhD Program FIS330, Statistical Mechanics.
Mar 2024 – Jul 2024	**Academic Support Mentor**, UTFSM. Course: FIS245, Mathematical Methods in Physics. *Provided specialized tutoring for a visually impaired student. Adapted course materials into accessible Markdown/LaTeX formats and designed tactile physical models for geometric visualizations.*
	Teaching Assistant, UTFSM. Course: PhD Program FIS330, Statistical Mechanics.
Mar 2021 – Jul 2021	**Teaching Assistant**, UTFSM. Course: General Physics I (FIS110), Physics & Astrophysics BSc program.

Languages

Spanish	Native speaker. Full professional proficiency in speaking, reading, and writing.
English	Fluent (C1). Full professional proficiency.
German	Upper-intermediate (B2). Certified by *Deutsches Sprachdiplom der Kultusministerkonferenz* (DSD II).
Chinese	Basic (A1). HSK-1 level.

References

Prof. Patricio Vargas
Full Professor of Theoretical Physics & Director of the PhD Program in Physics, UTFSM
Research areas: Magnetism, Electronic Properties of Modern Materials, Nanotechnology & Medical Physics
patricio.vargas@usm.cl

Dr. Francisco J. Peña
Academic Researcher / Assistant Professor, USS
PhD in Physical Sciences (PUCV, 2015).
Research areas: Quantum Thermodynamics and Energy Transfer in Nanosystems.
francisco.pena@usm.cl